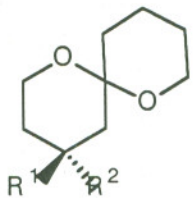
CUMULATIVE EXAM
ORGANIC CHEMISTRY

Dept. of Chemistry and Biochemistry
2/13/93

Acetals and ketals are a very important set of functional groups within organic chemistry. Structure 1 is an example of a spiroketal, a molecule containing a ketal unit within two rings which are connected at only one carbon atom. As with all acetals and ketals, this unit is formally derived from the dehydration of a carbonyl unit and two alcohol units.



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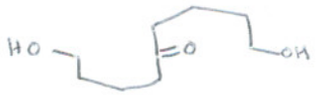
1. Draw the molecule which on dehydration would give spiroketal 1. (10pts)

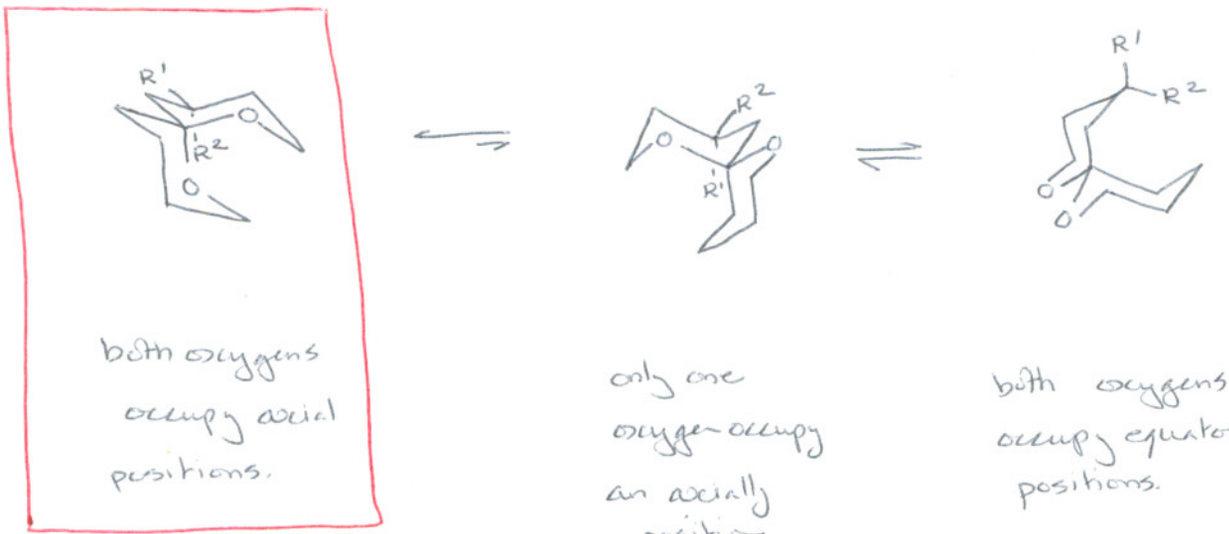
2. Using your knowledge of the anomeric effect (remember sugar chemistry) predict and draw in 3-dimensions the shape and orientation of spiroketal 1 (R^1, R^2 = H). Since most of us can not draw as well as Michelangelo, describe the important features of this structure and how they control the shape of the molecule. (40pts)

3. If R^1 and R^2 = Me, is the stereogenic spiro center of an R or an S configuration? (15pts)

4. If R^1 = H and R^2 = t-Bu what conformation would the molecule adopt (draw the structure and describe the important features) (35pts)

Organic Cume!

① 

② 

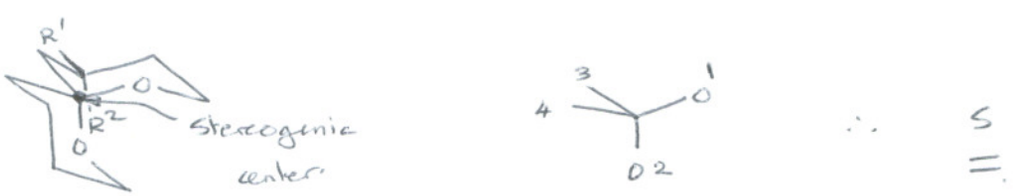
both rings are chairs - most stable

anomeric effects put oxygens axial.

(overlap of oxygen lone pairs with C-O σ* orbitals,

see Carey + Sundberg "Advanced Organic Chemistry, A"

pp 146-157.).

③ 

④ if $R^2 = Bu^t$ this put a large bulky group in an axial orientation (it is unhappy). The molecule isomerizes to.

$R^2 = t\text{-}Bu$ (one ring flip, and inversion of the anomeric center).